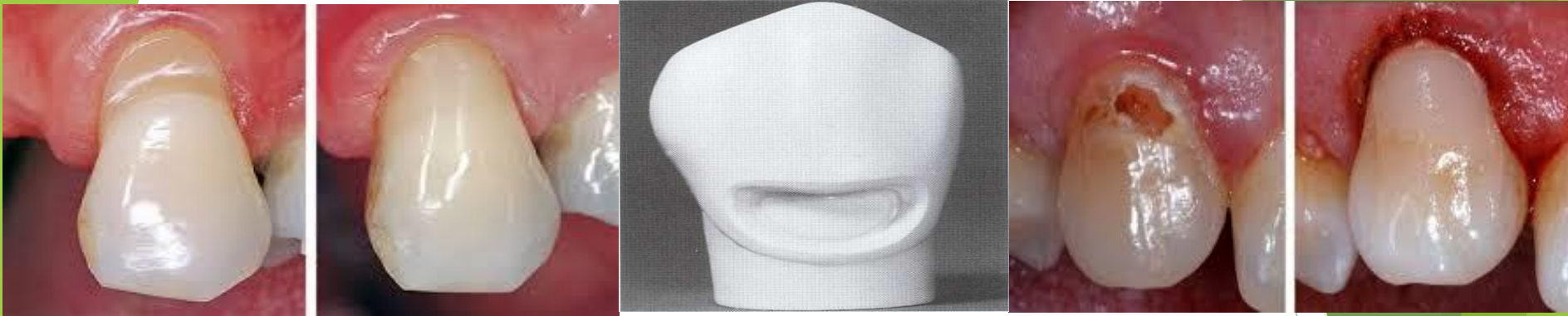




CLASS V CAVITY PREPARATION and RESTORATION

INTENDED learning outcomes

By the end of lecture, the student should be able to,

- Describe the cavity design for class V cavity preparation
- Explain the restoration steps and technique for class V cavity preparation

Definition

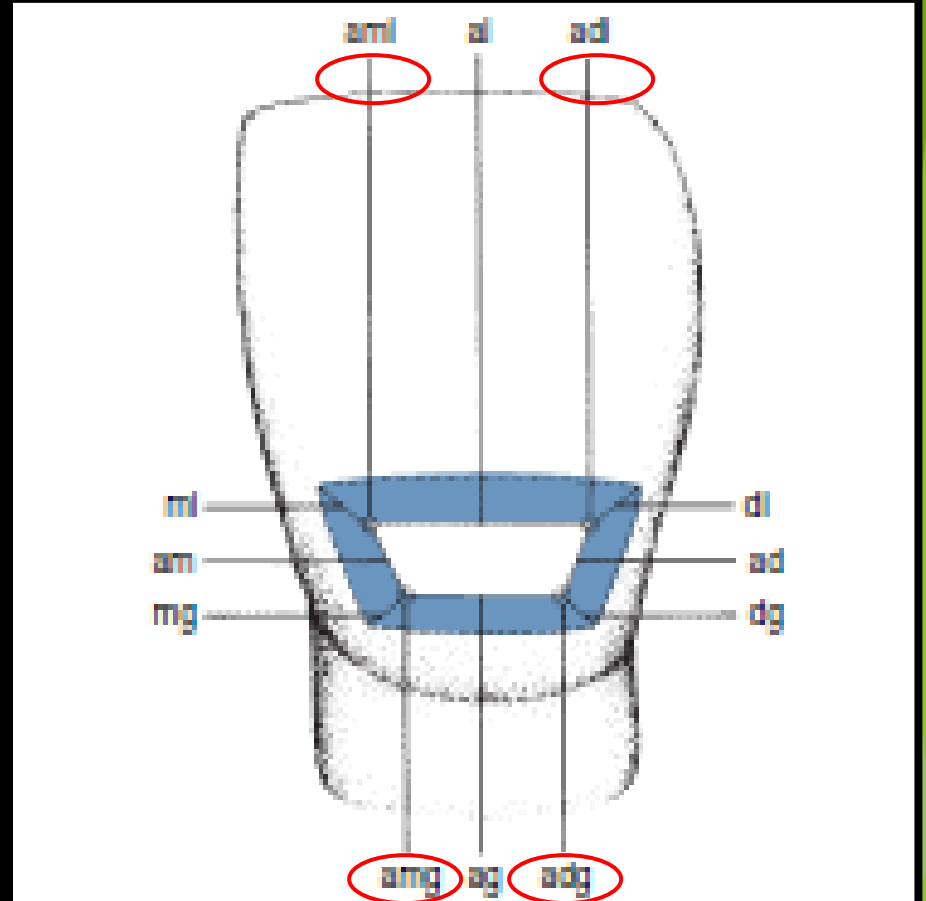
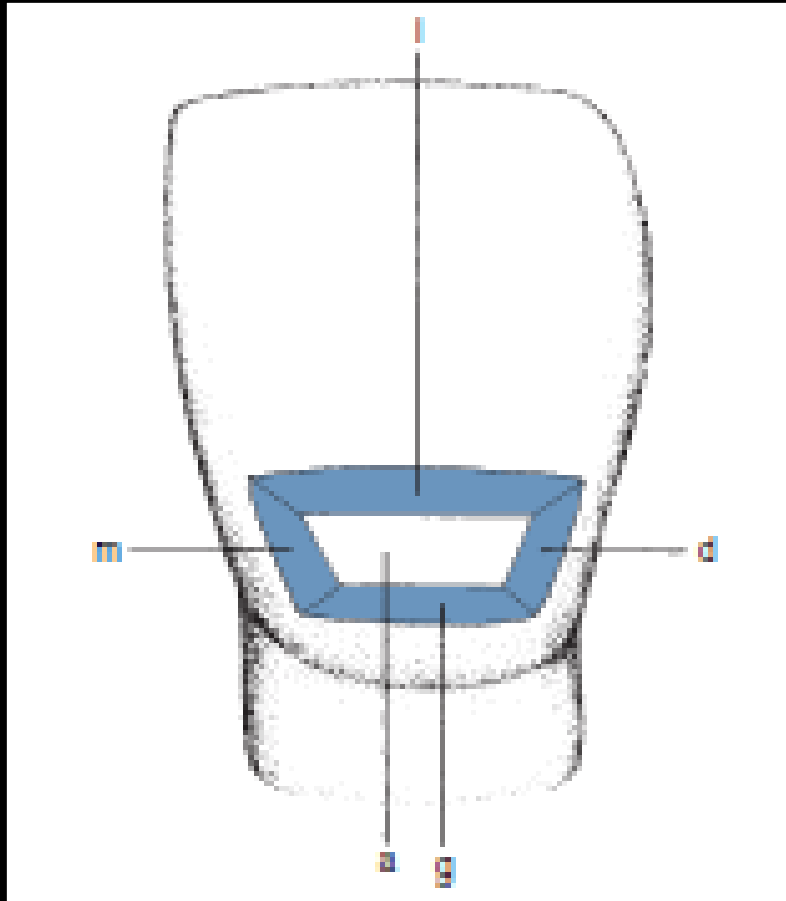
Class V Preparation:

Preparation on the gingival third of the facial or lingual surface of all teeth.

- Simple Preparation
- May be compound preparation (when involve mesial or distal surface)



(5) Walls, (8) Line angles and (4) point angles in class V cavity Preparation



For posterior teeth : - Incisal becomes Occlusal

Material selection

- ▶ Because of esthetic considerations, composites most frequently are used for the restoration of cervical caries lesions in anterior and premolar teeth.
- ▶ Microfilled composites may be selected for restoring Class V defects because their composition results in
 - ▶ (1) increased restoration smoothness and
 - ▶ (2) restoration flexibility when the tooth undergoes cervical flexure, especially in abfraction non carious lesions

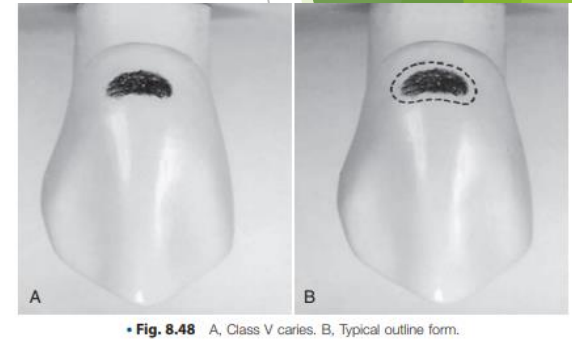


- ▶ Other factors like caries activity, caries risk, access to the lesion, decrease salivary function and moisture control must be taken into consideration in material selection.
- ▶ Resin modified glass ionomer cement and compomer can be used with patients with high caries risk and caries lesion extending on root surface

Preparation design

Similar to the Class III and IV preparations, tooth preparation for a Class V direct composite restoration involves

- (1) creating access to the defect (caries lesion, noncarious defect)
- (2) removing the defect (caries lesion, defective dentin and enamel, defective restoration and base material),
- (3) Making a convenience form for the restoration.



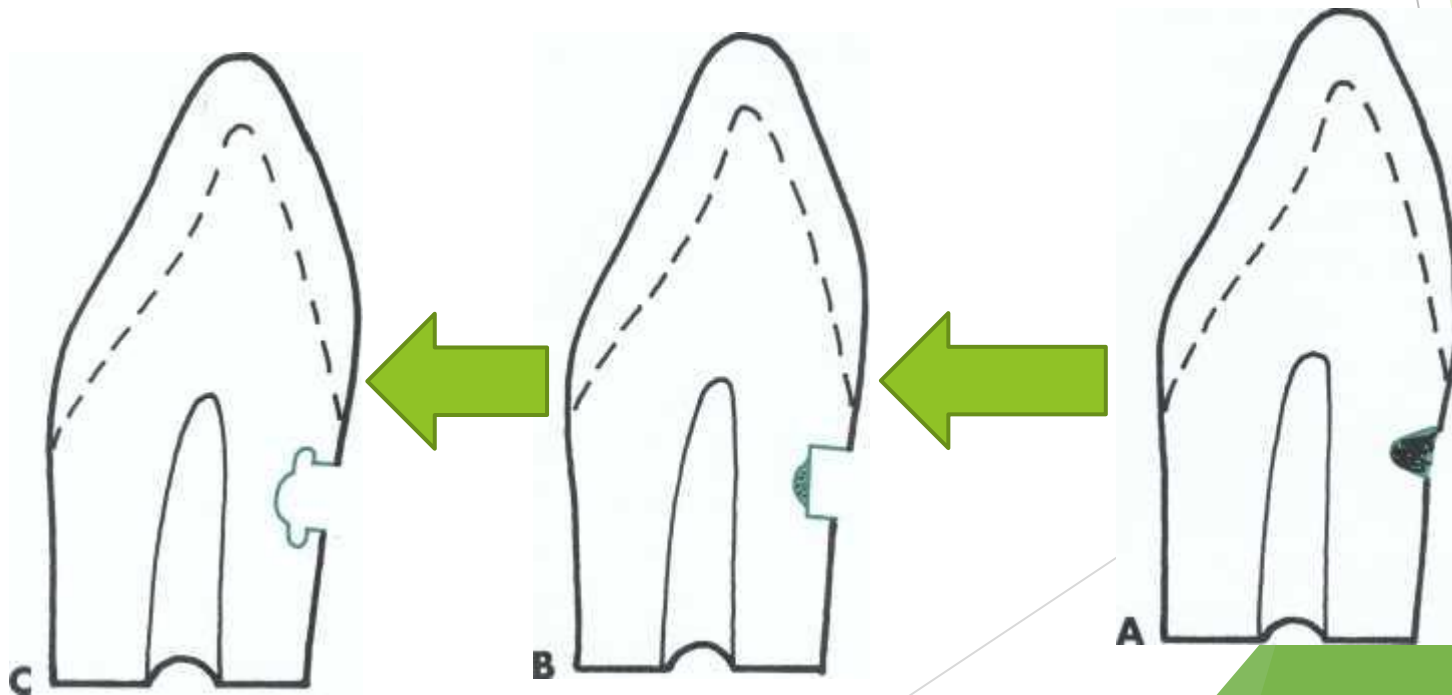
- ▶ The objective of the Class V tooth preparation for small or moderate lesions or defects that do not extend onto the root surface is to restore the lesion or defect as conservatively as possible.
- ▶ The preparation is extended into dentin only when the defect warrants such extension.

3 types of cavity designs

- ▶ 1.CONVENTIONAL
- ▶ 2.BEVELLED CONVENTIONAL
- ▶ 3.MODIFIED

I. Conventional:

Indicated for lesions which are partially or totally located on root surface
tooth preparation for class v composites



Important features :

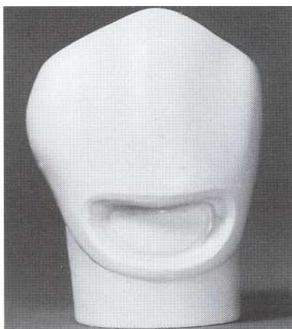
- 90 degree butt joint cavosurface margin on root surface
- When the cavity preparation is in the root surface, axial depth is 0.75 mm depth and retentive groove is sometimes indicated for retention
- Groove is done with $\frac{1}{4}$ round bur



II. Bevelled Conventional:



- This design is indicated when the entire large carious lesion is in enamel
- Also indicated for the replacement of an existing, defective Class V restoration.



- Using a **round bur** of suitable size, enter the carious lesion, removing the entire carious enamel lesion or defect.
- No effort is made to prepare 90 degree Cavo surface margins.
- Extend the preparation incisally, gingivally, mesially, and distally until the **cavosurface margins are positioned in sound tooth structure** providing the desired outline form

- Because the axial wall follows the mesiodistal convexity , incisogingival contours of the facial surface of the tooth will be usually convex.
- If carious tissue remains in dentin, it is selectively removed with a low-speed round bur or spoon excavator.

- Enamel bevels are typically used on the occlusal/incisal margins of the preparation, while at the cervical margin, bevels are usually not recommended because of the absence of enamel in this area.
- Bevelling is done with tapered flame shaped bur at 45 degree to the enamel margin



- **Function of bevel:**

- ❖ Increases the surface area for acid etching and retention
- ❖ Reduced microleakage owing to the enhanced bond between the composite and the tooth
- ❖ Beveling helps the restoration to blend more esthetically with the color of the surrounding tooth structure.

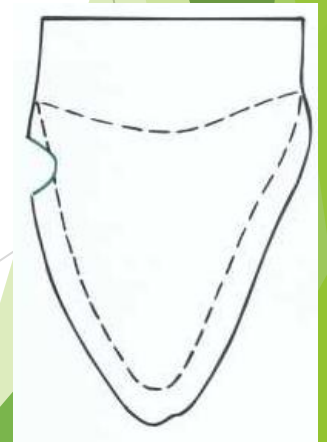


Variation of
bevelled
conventional
design with **wing
like extension** to
include
demineralised area
in outline form



► III. Modified:

- This design is indicated when Class-V caries lesions are very small cavitated lesions which are entirely on enamel, small hypoplastic enamel defects or non carious lesions
- The objective is to restore as conservatively as possible
- The lesion or defect is “scooped” out, resulting in a preparation form that may have a divergent wall configuration and an axial surface that usually is not uniform in depth



□ Proper isolation by :

(1) Cotton roll isolation and retraction cord (or)

(2) Rubber dam and suitable cervical retainer (No. 212)

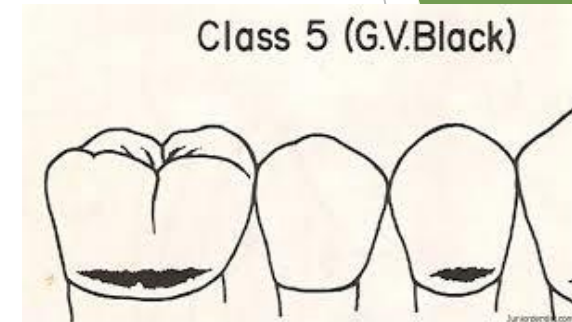


Restorative materials for class v cavity preparation

1. Composite
2. Resin modified Glass Ionomer cement
3. Compomer

INDICATIONS for composite

- All the cervical carious lesions whether
 - ❖ Simple or Compound
 - ❖ Anterior or Posterior
 - ❖ Labial/facial or Lingual
- Non-carious cervical lesions like abrasion and abfraction



CONTRAINDICATIONS for composite

- Sub gingival area/ root surface area
- Poor oral hygiene
- High caries risk
- Habits (bruxism)

ADVANTAGES of class v composite

- ✓ Esthetics
- ✓ Bonded to tooth structure
- ✓ Conservation of the tooth structure removal
- ✓ Less complex tooth preparation.
- ✓ Repairable

DISADVANTAGES

- ✓ Technique sensitive
- ✓ Polymerization shrinkage
- ✓ Less wear resistance
- ✓ More expensive

RESTORATIVE TECHNIQUE

- Acid etching and placement of the adhesive.
- Insertion and Light – activation of the composite.
- Contouring and polishing of the composite.

NOTE: Usually, no matrix is needed for class V restorations because the contour can be controlled as the composite restorative material is being inserted.



Cervical Matrix

- ▶ Specially designed for Class V restorations
- ▶ Easily adapted to all class V restorations
- ▶ Allows light cure composites through the matrix while it is held under pressure
- ▶ Obtains an excellent tooth anatomy reducing the chair time
- ▶ Providing a smooth surface that will require less adjustments



A flame-shaped finishing bur or diamond is recommended for removing excess composite on the facial surface of a Class V composite



Abrasive discs, mounted on an appropriate mandrel in an angle handpiece at low speed can be used. The external enamel surface should act as a guide for proper contour.

A constant shifting motion aids in contouring and preventing the development of a flat surface.

A fine-grit disc is used for final polishing.



Rotary instruments should be used carefully in gingival locations (especially on the root surface) to prevent inadvertent and undesirable removal of tooth structure (usually cementum and dentin)



Reference text book

- ▶ Sturdevant's Art and Science of Operative Dentistry - 5th and 6th Edition
- ▶ Chapter 12- pages 556-562
- ▶ Chapter 18- pages 786 and 796-805

THANK YOU